



Smart Jacket For Social Distancing

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This smart social distancing jacket comes with a smart inbuilt alarm, hinting people around you to maintain distance in crowded places

like malls, offices, schools, colleges, and hospitals. Here is its novel circuit that alerts people to maintain physical distance all around you. The author's

prototype with a jacket fitted with sensors is shown in Fig. 1.

Circuit and working

Circuit diagram of the smart jacket for social distancing is shown in Fig. 2. It is built around Arduino Uno Board1, three HC-SR04 ultrasonic sensors US1 through US3, and piezo buzzer PZ1.

The three HC-SR04 ultrasonic sensors are used to detect the distance from the back, left, and right sides of the user. Piezo buzzer PZ1 sounds alarm when any of the sensors detects a person within 30 to 100cm distance.

The Arduino board can be fixed at the jacket's back side along with the buzzer. The inbuilt LED of Arduino also notifies the person standing at the back side as shown in Fig. 3.

The software program *smart_jacket.ino* loaded into internal memory of the Arduino is simple and easy to understand. Comments are given at the end of each command line. The code starts by defining the pin numbers of sensors. The trigger pin (input) and echo pin (output) of each sensor are defined in the code.



Fig. 1: Author's prototype

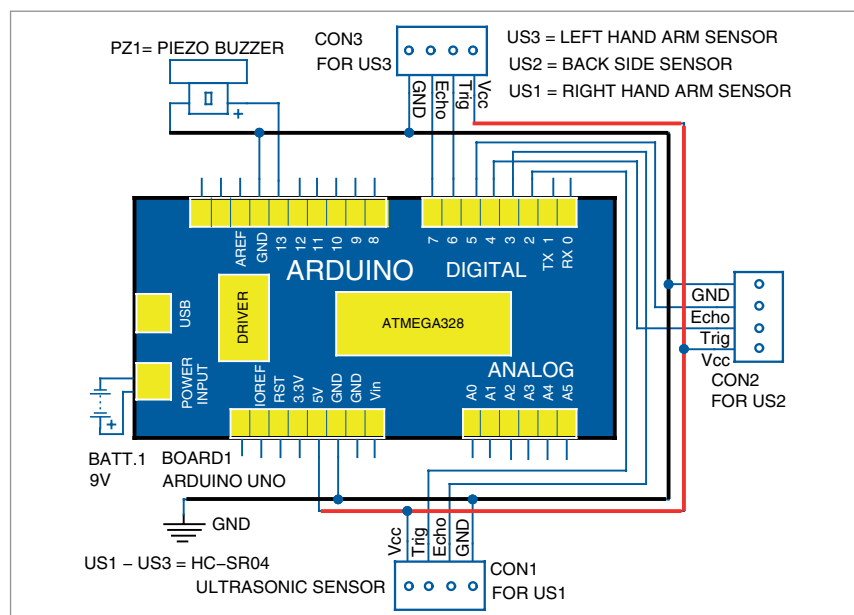


Fig. 2: Circuit diagram



Fig. 3: Sensor and Arduino on jacket's rear side